



BEFORE YOU READ – 3 BLIND ID CHALLENGES

1. Which region has Brunner's glands, which has Peyer's patches, and which has the most plicae circulares?
2. List the four ways the small intestine increases its absorptive surface.
3. Name the six cell types of the crypts and villi — which one carries lysozyme granules?

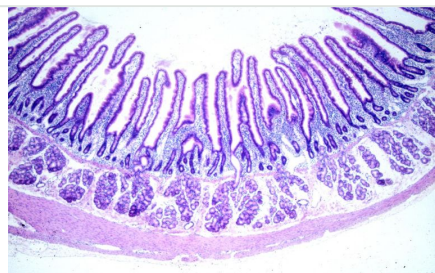
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 · Three regions — one-line ID rules

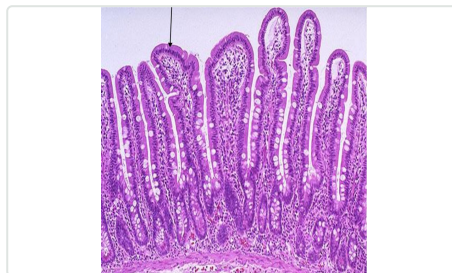
	DUODENUM	JEJUNUM	ILEUM
Length	25 cm	2.5 m	3.5 m
Villi	leaf-shaped	long, finger-shaped	fewer, club-shaped
Goblet cells	few	intermediate number	abundant
The ID feature	Brunner's glands in the submucosa; bile + pancreatic duct open into this C-shaped segment	many plicae circulares	Peyer's patches (lymphoid nodules) in the lamina propria
Outer covering	serosa & adventitia	serosa	serosa

Brunner's glands secrete **alkaline mucus**: it protects the duodenal lining from the acidity of the chyme and brings the contents to the optimum pH for pancreatic enzyme action. The small intestine's jobs are **absorption** and **secretion of enzymes**; it ends at the **ileocecal valve**, a sphincter controlling entry into the large intestine.

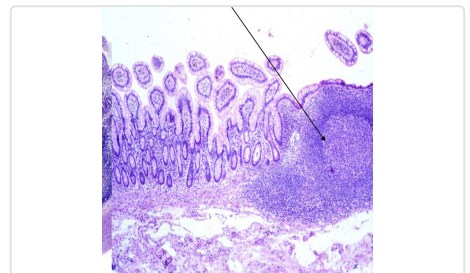
Slide wording: the lecture calls the small intestine "approximately 5 m", but its own regional lengths add up to about 6.25 m — quote the regional lengths.



Duodenum: leaf-shaped villi over submucosal Brunner's glands.



Jejunum: long finger-shaped villi.

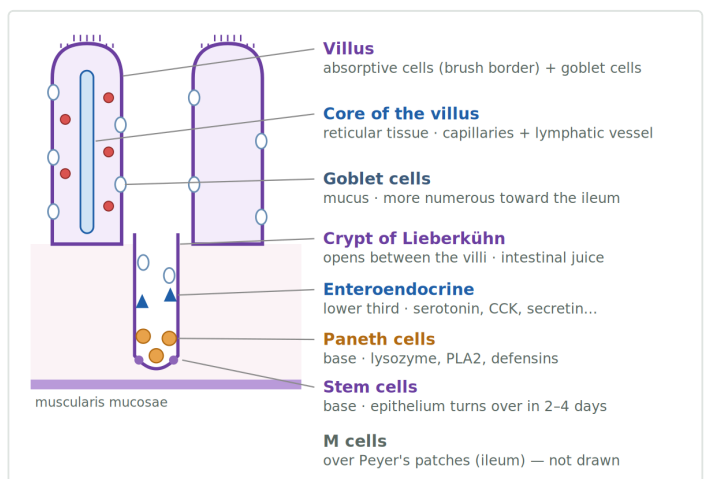


Ileum: club-shaped villi; Peyer's patch (arrow).

2 · Making absorptive surface — and the crypt-villus unit

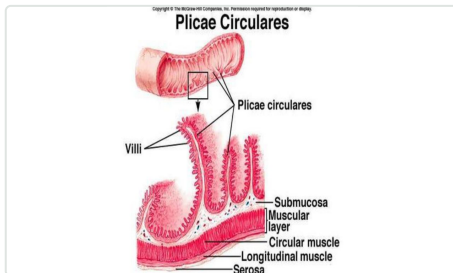
	MECHANISM	WHAT IT IS
1	Length	long tube → prolonged contact between food and enzymes
2	Plicae circulares	folds of mucosa + submucosa ; found in the jejunum; slow the passage of contents
3	Villi	finger-like mucosal projections; reticular-tissue core with blood capillaries + a lymphatic vessel, covered by epithelium
4	Microvilli	surface cytoplasmic extensions = the brush border

- **Crypts of Lieberkühn** = simple tubular glands that extend into the lamina propria below the base of the villi; their cells secrete **intestinal juice**, entering the lumen through small openings between the villi.

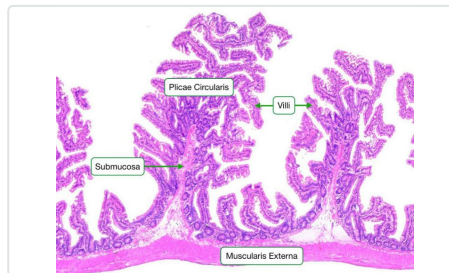


Anchor diagram: villus and crypt — which cell sits where.

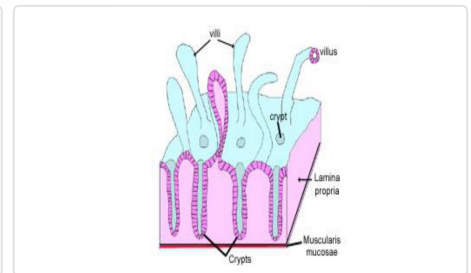
MAG	WHAT TO LOOK FOR
×4	Wall layers and plicae circulares (mucosa + submucosa folds, jejunum); villi fringe the lumen; Peyer's patches bulge in the ileum.
×10	Villus shape (leaf / finger / club) and the crypts at the base; Brunner's glands in the submucosa = duodenum.
×40	Brush border on tall columnar cells with basal oval nuclei; goblet cells ; Paneth granules at the crypt base.



Plicae circulares, villi, submucosa, muscle layers, serosa.



Labelled: plicae circularis, villi, submucosa, muscularis externa.

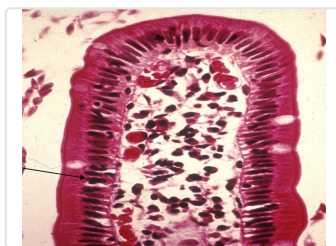


Villi, crypts, lamina propria, muscularis mucosae.

3 · Six epithelial cell types – the grid

CELL	WHERE	ID ON THE SLIDE	PRODUCT / JOB
Absorptive	villus surface and crypts	tall columnar , oval basal nucleus, brush border (~3,000 microvilli per cell)	absorption; microvilli increase surface area
Goblet	between absorptive cells; increase toward the ileum	interspersed mucus-filled cells	mucus for protection and lubrication
Paneth	basal part of the crypts	exocrine; basophilic cytoplasm, eosinophilic apical granules	lysozyme, phospholipase A2, defensins break down microbial membranes and bacterial cell walls; regulate the crypt microenvironment
M (microfold)	overlying Peyer's patches (ileum)	specialised epithelial cell; discontinuous basement membrane	endocytose antigens and pass them to underlying macrophages / lymphocytes (protective)
Enteroendocrine	lower third of the crypt	triangular ; resemble stomach APUD cells	serotonin, somatostatin, cholecystokinin, peptides, secretin → motility, peristalsis, secretions, visceral sensation, appetite
Stem cell	base of the crypt	undergo mitosis	renew the epithelium; most GI epithelial cells turn over in 2-4 days

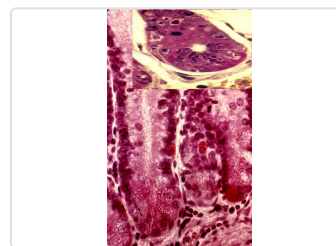
Look-alike pair: Paneth and enteroendocrine cells both sit low in the crypt. **Paneth = eosinophilic granules at the apex (lysozyme, antimicrobial); enteroendocrine = triangular hormone-secreting cell.**



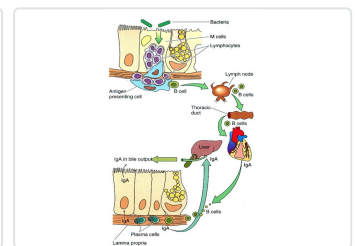
Absorptive cells: brush border.



Goblet cell among absorptive cells.



Paneth cells at the crypt base.



M cell over a lymphoid follicle: antigen → lymphocytes.

THE REST OF THE WALL

- Lamina propria:** loose CT with blood and lymph vessels, nerves, smooth muscle, lymphoid cells and macrophages. **Muscularis mucosae:** typical.
- Submucosa:** dense CT (contains **Brunner's glands** in the duodenum). **Muscularis externa:** 2 layers — inner circular, outer longitudinal. Then serosa or adventitia.

Pearls — what must survive if everything else is forgotten

- **Region ID:** Brunner's glands + leaf villi = duodenum; long finger villi + many plicae circulares = jejunum; club villi + abundant goblet cells + Peyer's patches = ileum.
- **Surface area:** length → plicae circulares (mucosa + submucosa) → villi (capillaries + lymphatic vessel) → microvilli (~3,000 per cell).
- **Paneth cells** (crypt base): lysozyme, phospholipase A2, defensins. **Enteroendocrine** (lower third, triangular): serotonin, somatostatin, CCK, secretin. **Stem cells** (base): turnover 2–4 days.
- **M cells** overlie Peyer's patches, have a discontinuous basement membrane and endocytose antigens to macrophages / lymphocytes.
- **Brunner's glands** secrete alkaline mucus: protects the duodenum from acid chyme and sets the pH for pancreatic enzymes.

Test yourself — transfer-task MCQs

Unseen fields, not slide recall. Commit to an answer before you uncover the key.

1. A duodenal biopsy shows glands in the submucosa. What is the main job of their secretion?

- A. Digest protein using pepsin
- B. Protect the lining from acid chyme and bring contents to the optimum pH for pancreatic enzymes
- C. Kill bacteria with lysozyme and defensins
- D. Transport antigens to lymphocytes
- E. Secrete intrinsic factor

2. A section shows club-shaped villi, abundant goblet cells and lymphoid nodules in the lamina propria. Which segment?

- A. Duodenum
- B. Jejunum
- C. Ileum
- D. Appendix
- E. Pylorus

3. An epithelial cell overlies a lymphoid follicle in the ileum, has a discontinuous basement membrane and takes up antigens by endocytosis. What is it?

- A. Paneth cell
- B. Goblet cell
- C. Enteroendocrine cell
- D. Stem cell
- E. M cell

4. A cell at the crypt base has basophilic cytoplasm and eosinophilic apical granules. Which set does it release?

- A. Serotonin and secretin
- B. Mucin
- C. Lysozyme, phospholipase A2 and defensins
- D. Pepsinogen and gastric lipase
- E. Cholecystokinin and somatostatin

5. Which list gives the four ways the small intestine increases its absorptive surface?

- A. Villi, crypts, Brunner's glands, Peyer's patches
- B. Length, plicae circulares, villi, microvilli
- C. Plicae circulares, goblet cells, Paneth cells, microvilli
- D. Length, taeniae coli, villi, microvilli
- E. Villi, microvilli, rugae, plicae semilunares

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. **B** — Brunner's glands secrete alkaline mucus, protecting the duodenal lining and setting the pH for pancreatic enzyme action.
- 2. **C** — Fewer club-shaped villi, abundant goblet cells and Peyer's patches = ileum.
- 3. **E** — M (microfold) cells overlie Peyer's patches and pass antigens to underlying macrophages and lymphocytes.
- 4. **C** — Paneth cells release lysozyme, phospholipase A2 and defensins that break down microbial membranes and cell walls.
- 5. **B** — The lecture lists length, plicae circulares, villi and microvilli (crypts are secretory glands).

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 2, Dr Hero K. Mustafa
Study alongside the lecture, not instead of it.